



Two Kinds of AI

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Rule-Based Systems (like Truffle Hunt) vs. Large Language Models (like ChatGPT or Claude)

Rule-Based AI

As seen in: Truffle Hunt

EXPLICIT

HOW IT WORKS

- You write explicit IF-THEN rules.
- The system checks each rule against the current board state.
- If a pattern matches, the prescribed action is taken.
- If no rules match, the system does nothing.
- Every decision can be read and explained by a human.

Analogy:

A recipe book. Follow the steps exactly. If the ingredient isn't listed, stop.

STRENGTHS

- Fully transparent and explainable
- Predictable — same input, same output
- No training data needed
- Easy to debug and modify

LIMITS

- Can't handle situations no rule covers
- Rules must be written for every case
- Doesn't learn or improve on its own

Large Language Models

As seen in: ChatGPT, Claude, Gemini

LEARNED

HOW IT WORKS

- Trained on billions of words of text.
- Learns statistical patterns: which words tend to follow which.
- Generates responses by predicting the most fitting next token.
- No explicit rules — behavior emerges from training data.
- The same input may produce slightly different outputs each time.

Analogy:

A chef who has tasted thousands of dishes. No recipe needed — intuition guides what tastes right.

STRENGTHS

- Flexible — handles novel, open-ended questions
- Generalizes across many topics and domains
- Natural language in, natural language out
- Improves with more and better training data

LIMITS

- Opaque — hard to explain why it said what it said
- Can produce confident-sounding errors (hallucinations)
- Requires massive data and compute to train

	Rule-Based AI	Large Language Model
Knowledge	Explicit rules you write	Patterns from training data
Transparency	Fully explainable	Often a 'black box'
Flexibility	Only what rules cover	Open-ended and generative
Learning	Doesn't learn	Learns during training
Output	Deterministic	Probabilistic

Both are forms of AI — but they think very differently. In Truffle Hunt, you are the AI designer: you write the rules that guide your boar's reasoning.

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